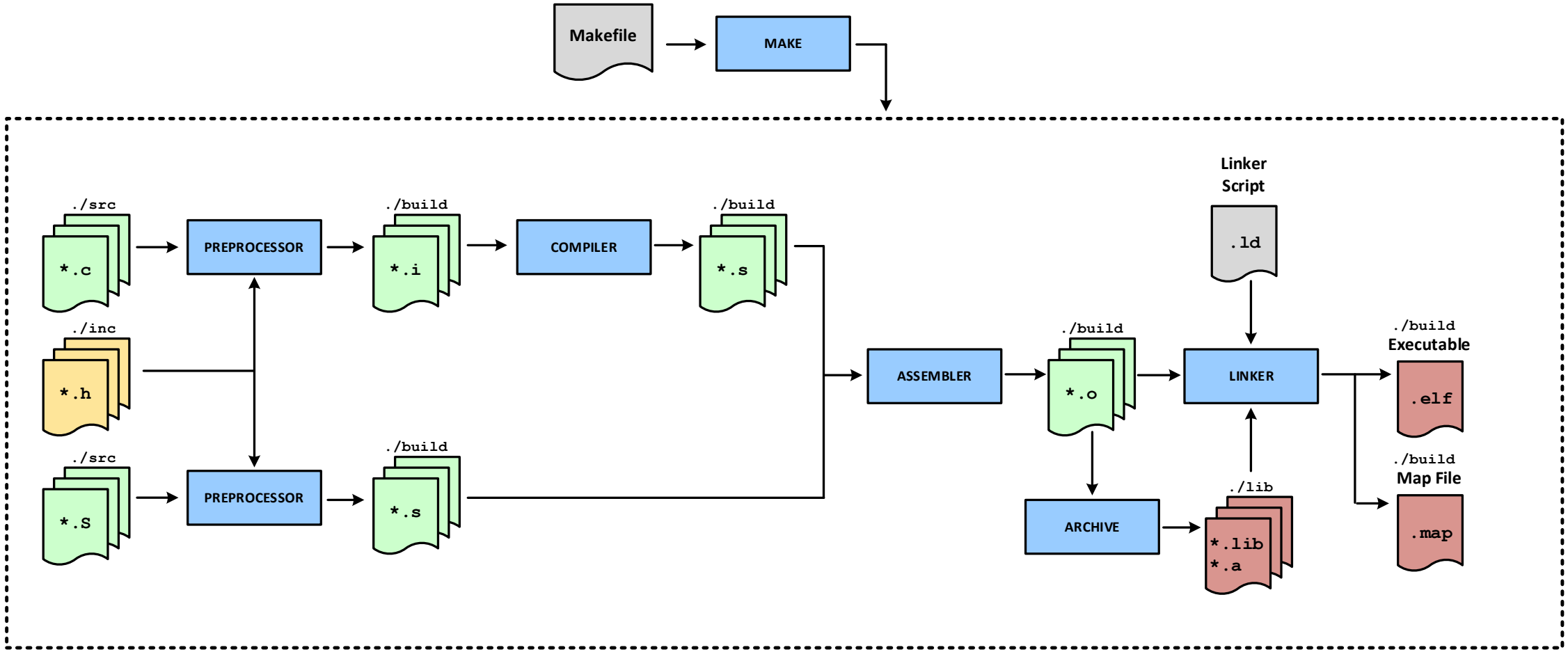
EE426 / 626 Microprocessor Applications

GNU Toolchain



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Source Files

```
% cd ~/embedded/projects/stm32
% vim src/main.c
% vim inc/main.h  (or 'gi')
```

VIM and LSP Bindings

,cd	netrw
gd	Go to definition (Ctrl-o and Ctrl-i to navigate)
gr	Find references
gh	Show hover
gp	Definition popup
gs	List symbols
,rn	Rename symbol



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Preprocessor

```
% mkdir build

% arm-none-eabi-gcc                                                          \
        -mcpu=cortex-m4 -mthumb -mfpu=fpv4-sp-d16 -mfloat-abi=hard           \
        -std=gnu11                                                           \
        -Iinc -Isegger -Idrivers/CMSIS/Include -Idrivers/CMSIS/Device/ST/STM32F3xx/Include  \
        -DSTM32F303xC                                                        \
        -E src/main.c -o build/main.i

% vim build/main.i
```



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Compiler

```
% arm-none-eabi-gcc                                                    \
        -mcpu=cortex-m4 -mthumb -mfpu=fpv4-sp-d16 -mfloat-abi=hard     \
        -Wall -Wextra -g3 -Og -std=gnu11 -ffunction-sections -fdata-sections  \
        -S build/main.i -o build/main.s

% vim build/main.s
```



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Assembler

```
% arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -mfpu=fpv4-sp-d16 -mfloat-abi=hard          \
        -c build/main.s                                                                 \
        -o build/main.o

% arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -mfpu=fpv4-sp-d16 -mfloat-abi=hard          \
        -x assembler-with-cpp                                                           \
        -c startup/startup_stm32f303cctx.s                                              \
        -o build/startup_stm32f303cctx.o

% arm-none-eabi-objdump -d build/main.o | less
```



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Archive

```
% arm-none-eabi-ar t $(arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -mfpu=fpv4-sp-d16          \
       -mfloat-abi=hard -print-file-name=libc.a)

% arm-none-eabi-ar rcs build/libdemo.a build/main.o build/startup_stm32f303cctx.o

% ls -la build/libdemo.a

% arm-none-eabi-ar t build/libdemo.a
```



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Linker

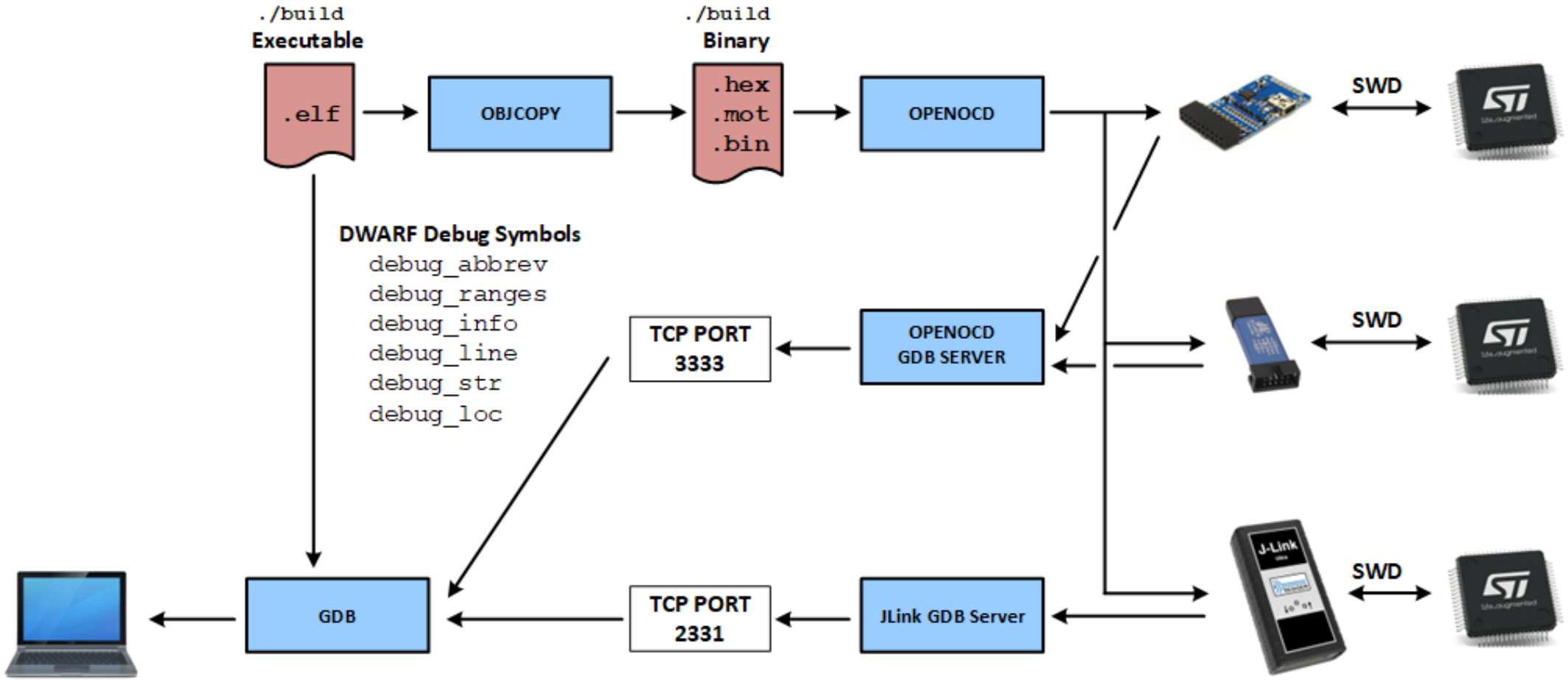
```
% arm-none-eabi-gcc                                                    \
    $(find build -name '*.o' ! -name 'main.o' ! -name 'startup_stm32f303cctx.o')  \
        -Lbuild                                                        \
        -ldemo                                                         \
        -mcpu=cortex-m4 -mthumb -mfpu=fpv4-sp-d16 -mfloat-abi=hard     \
        -TSTM32F303CCTX_FLASH.ld -Wl,-Map=build/clk.map                \
        -Wl,--gc-sections -Wl,--print-memory-usage                     \
        -specs=nano.specs -specs=nosys.specs -u _scanf_float           \
        -Wl,--start-group -lc -lm -Wl,--end-group                      \
        -o build/clk.elf

% arm-none-eabi-size build/clk.elf

% file build/clk.elf
```



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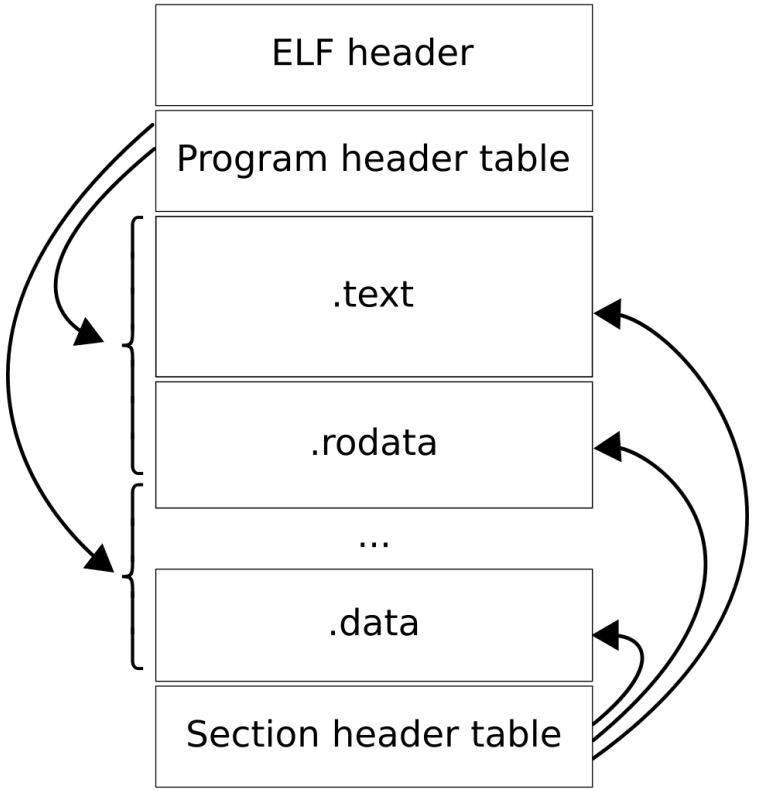


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ELF – Executable and Linkable Format

```
.elf   .so
.out   .ko
.bin   .o
.axf
```

```
[aturvey@euler:build]% (main *+) hexdump -C -n 64 clk.elf
00000000  7f 45 4c 46 01 01 01 00  00 00 00 00 00 00 00 00  |.ELF............|
00000010  02 00 28 00 01 00 00 00  55 0d 00 08 34 00 00 00  |..(.....U...4...|
00000020  78 82 0d 00 00 04 00 05  34 00 20 00 03 00 28 00  |x.......4. ...(.|
00000030  1c 00 1b 00 01 00 00 00  00 10 00 00 00 00 00 08  |................|
```



By Surueña - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=2922583

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OBJDUMP

```
% arm-none-eabi-readelf -h build/clk.elf          # ELF header — entry point, machine type
% arm-none-eabi-readelf -S build/clk.elf          # section headers
% arm-none-eabi-objdump -h build/clk.elf          # same section table, objdump's view
% arm-none-eabi-objdump -d build/clk.elf          # full disassembly
```



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LINKER SCRIPT

STM32F<>_FLASH.ld

SECTIONS

.isr_vector
.text
.data
.rodata
.bss
.user_heap_stack
.ccmram

```
/* Entry Point */
ENTRY(Reset_Handler)

/* Highest address of the user mode stack */
_estack = ORIGIN(RAM) + LENGTH(RAM); /* end of "RAM" Ram type memory */

_Min_Heap_Size = 0x200; /* required amount of heap */
_Min_Stack_Size = 0x400; /* required amount of stack */

/* Memories definition */
MEMORY
{
  CCMRAM    (xrw)    : ORIGIN = 0x10000000,   LENGTH = 8K
  RAM       (xrw)    : ORIGIN = 0x20000000,   LENGTH = 40K
  FLASH     (rx)     : ORIGIN = 0x08000000,   LENGTH = 256K
}

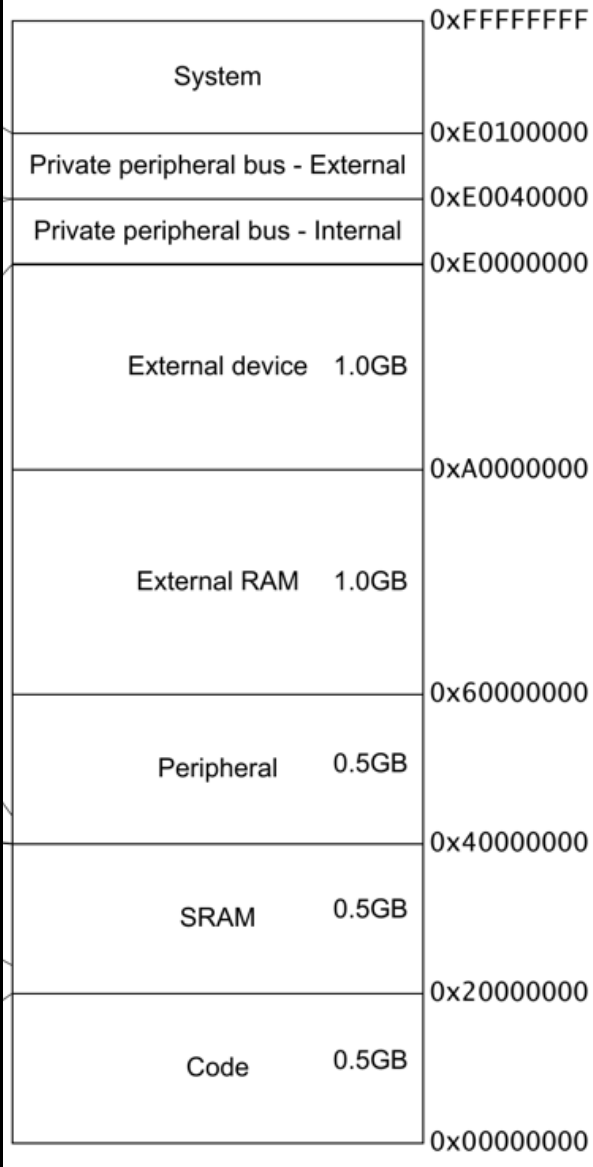
/* Sections */
SECTIONS
{

  /* The startup code into "FLASH" Rom type memory */
  .isr_vector :
  {
    . = ALIGN(4);
    KEEP(*(.isr_vector)) /* Startup code */
    . = ALIGN(4);
  } >FLASH

  /* The program code and other data into "FLASH" Rom type memory */
  .text :
  {
    . = ALIGN(4);
    *(.text)           /* .text sections (code) */
    *(.text*)          /* .text* sections (code) */
    *(.glue_7)         /* glue arm to thumb code */
    *(.glue_7t)        /* glue thumb to arm code */
    *(.eh_frame)

    KEEP (*(.init))
    KEEP (*(.fini))

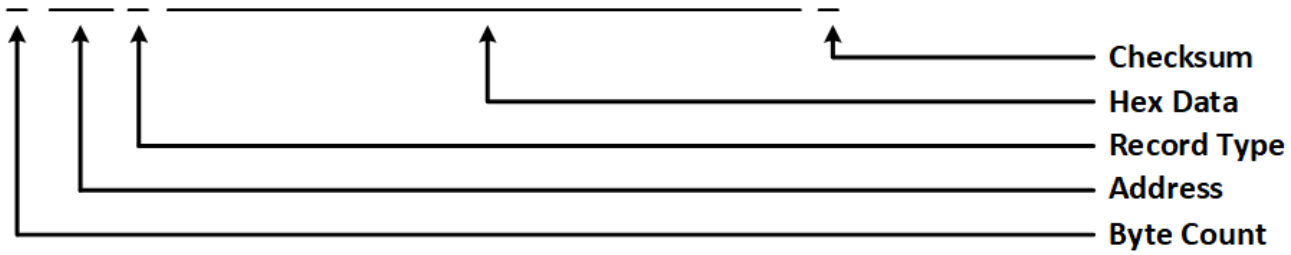
    . = ALIGN(4);
    _etext = .;        /* define a global symbols at end of code */
  } >FLASH
```



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Intel HEX Format

```
:02000004080F2
:10000000A00020550D0008A50D0008A50D000852
:10001000A50D0008A50D0008A50D000800000B2
:10002000000000000000000000000A50D000816
:10003000A50D00080000000A50D00083D4D0008BA
:10004000A50D0008A50D0008A50D00082141000818
:10005000A50D0008A50D0008A50D0008A50D0008B8
```



Record Types (for EE426/626)

00 Data

```
:04001000FFFFFFFFFE
```

01 End of File

```
:00000001FF
```

04 Extended Linear Address

```
:02000004040F6
```

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OBJCOPY

```
% arm-none-eabi-objcopy -O ihex build/clk.elf build/clk.hex

% arm-none-eabi-objcopy -O binary build/clk.elf build/clk.bin

% arm-none-eabi-objcopy -O srec   build/clk.elf build/clk.mot

% ls -la build/clk.*

% vim build/clk.hex
```



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FLASH

```
% JLinkExe -device STM32F303CC -if SWD -speed 4000 <<EOF
    > connect
    > halt
    > loadfile build/clk.elf
    > r
    > g
    > q
    > EOF
```



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GDB SERVER / GDB / TTY

```
% JLinkGDBServer -device STM32F303CC -if SWD -speed 4000 -port 2331

% arm-none-eabi-gdb -q -x .gdbinit-stm32f303cc build/clk.elf

% ll /dev/tty.usbserial-*
% picocom -b 115200 /dev/tty.usbserial-FTYQPTSO
      (ctrl-a + ctrl-x to exit)
```



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DEBUG

```
continue
ctrl-c
info reg
print $r0
info locals
info variables
print g_state_disp
print SystemCoreClock

list src/systick.c:27
list 0x08000d55
list *$pc

disassemble
```

```
info b
break SysTick_Handler
watch g_millis
continue
info b
delete <#>

break state_set_disp
watch g_state_disp
continue
info b
delete <#>

continue
```

```
set disassemble-next-line off
next (n)
step (s)


set disassemble-next-line on
step
next
stepi (si)
nexti (ni)
```



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MAKE

```
% vim Makefile

% make clean
% make compile

% make build

% make flash

% make serv
% make gdb
% make tty
```

GIT

```
% git status

% git log –pretty=oneline –n 1

% git add

% git commit –m ‘comment’

% git push

% git logp
```



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STM32CubeIDE

